



NUI Galway
OÉ Gaillimh

Semester II Examinations 2018/2019

Exam Codes	3BME1, 4BME1, 1EM1, 1OA2, 4BCT1, 4BS2, 4BMS2, 3BA1
Exams	3rd and 4th year (Arts and Science)
Module	Cryptography
Module Codes	CS402, MA492
External Examiner	Prof. T. Brady
Internal Examiners	Prof. G. Ellis Dr T. Rossmann*

Instructions Answer all four questions.
Each question carries 25 marks.

Duration	2 hours
No. of Pages	3 pages, including this one
School	Mathematics, Statistics and Applied Mathematics

Requirements: No special requirements

Release to Library	Yes
MCQ	No
Statistical / Log Tables	No
Release to Library	Yes
Non-programmable Calculator	Yes
Mathematical Tables	No

Q1. (a) The ciphertext string

ASISENGEHFGNONFVWTGNQWCVSE

was produced using a shift cipher over the 27-letter alphabet

$$A = 0, B = 1, \dots, Z = 25, _ = 26,$$

where the final character $_$ indicates a space.

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol **E** corresponds to the ciphertext symbol **S**. Determine the first word of the plaintext string corresponding to the ciphertext string given above. [10 marks]

- (b) Identify $\mathbb{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, _ \}$ as in part (a). An affine cipher over $\mathbb{Z}_{27}$ with unknown key $(a, b) \in \mathbb{Z}_{27}^\times \times \mathbb{Z}_{27}$ and encryption function

$$\mathbb{Z}_{27} \rightarrow \mathbb{Z}_{27}, \quad x \mapsto ax + b$$

encrypts **E** as **T** and **S** as **V**. Determine (a, b) . [5 marks]

- (c) State *Kerckhoff's principle*. [5 marks]

- (d) An affine cipher (of dimension 1) over $\mathbb{Z}_{55}$ is used to encrypt messages. Find the number of possible keys. [5 marks]

Q2. (a) State (without proof) *Shannon's Theorem* on perfectly secure cryptosystems. [5 marks]

- (b) Briefly describe *block ciphers* and *stream ciphers*. [5 marks]

- (c) Explain what is meant by a *linear feedback shift register* of length L . [5 marks]

- (d) A linear feedback shift register of length 3 with connection polynomial

$$C(X) = 1 + X + X^3$$

over $\mathbb{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = 0$, $s_1 = 0$, and $s_2 = 1$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$. [10 marks]

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- Q3.** (a) Briefly describe and contrast *symmetric cryptography* and *public-key cryptography*. [5 marks]
- (b) Describe the key generation step of the *RSA* cryptosystem. [5 marks]
- (c) Alice and Bob use the cyclic group $\mathbb{Z}_{25}^\times$ and its generator $g = 2$ to perform a *Diffie-Hellman key exchange*. Alice secretly chooses $a = 6$ and Bob secretly chooses $b = 3$. Determine their shared key (as an element of $\mathbb{Z}_{25}$). [5 marks]
- (d) Alice encrypts the message $m = 23$ using the *RSA* cryptosystem and Bob's public key $(n, e) = (77, 17)$. Find the corresponding ciphertext (as an element of $\mathbb{Z}_{77}$). [10 marks]

- Q4.** (a) Define *Euler's totient function* and state (without proof) *Euler's Totient Theorem*. [5 marks]
- (b) Describe the *Fermat primality test*. [5 marks]
- (c) Define *Carmichael numbers* and explain their role in primality testing. [5 marks]
- (d) Let E be the set of points of an elliptic curve over $\mathbb{Q}$ defined by the equation

$$y^2 = x^3 + ax + b.$$

Describe (either geometrically or algebraically) the usual operation of addition on E which turns E into an abelian group. [10 marks]